

Lag-Order Uncertainty in Set-Identified SVARs

A comparative presentation of two Monte Carlo studies

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Agenda

Introduction

Theory

Simulation Design

Results

- Paper I: BVAR shrinkage

- Paper II: Bayesian model averaging

Conclusion

The central problem

- ▶ Structural VARs are widely used for macroeconomic inference.
- ▶ Impulse responses depend strongly on the estimated lag structure.
- ▶ In finite samples, lag-order choices are uncertain and can be highly influential.
- ▶ Set-identified SVARs add another layer of uncertainty through the identification step.

Research question

Which estimation strategy recovers the true impulse response function most reliably when the lag order is uncertain?

Theory: overview

- ▶ VAR framework and impulse response functions.
- ▶ Set identification via sign restrictions and the median target.
- ▶ Information criteria and the bias–variance trade-off.
- ▶ Bayesian extension: shrinkage (Paper I) and model averaging (Paper II).

From VAR to impulse responses

A reduced-form VAR explains today by its own past:

$$Y_t = v + A_1 Y_{t-1} + \dots + A_{p_0} Y_{t-p_0} + e_t, \quad \mathbb{E}[e_t e_t'] = \Sigma$$

Inverting it (infinite MA form) rewrites today as accumulated past shocks:

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i e_{t-i}$$

- ▶ The **IRF** is the sequence Φ_i : a shock's echo over time.
- ▶ **Channel 1 – lag order**: Φ_i comes from $A_1, \dots, A_{p_0}$, so the number of lags shapes the IRF.
- ▶ **Channel 2 – identification**: e_t mixes structural shocks; isolating one needs $e_t = B\varepsilon_t$ with $\Sigma = BB'$.

Takeaway

An IRF depends on *two* choices. We target lag uncertainty; set identification adds a second layer (next).

Sign restrictions: identification without a single answer

The problem: infinitely many B satisfy $\Sigma = BB'$ — so which structural story is right?

- ▶ **Cholesky:** imposes a strong assumption (a fixed ordering) to pin down *one* answer.
- ▶ **Uhlig (2005) – agnostic:** let the data speak; impose only a few sign restrictions everyone can agree on.
- ▶ Sign restrictions rule out the “wrong-signed” models and keep the rest $\Rightarrow$ they identify a **set** of admissible IRFs, not one.

Picking one representative

To compare against a single true IRF, we take the **median target** (Fry & Pagan, 2011): the actual admissible model closest to the pointwise median.

The bias–variance trade-off

An information criterion embodies the trade-off:

$$\text{IC}(p) = \underbrace{\ln |\hat{\Sigma}(p)|}_{\text{penalizes underfitting}} + \underbrace{c_T \frac{K^2 p}{N}}_{\text{penalizes overfitting}} \quad (\text{MSE} = \text{Bias}^2 + \text{Variance})$$

- ▶ Too few lags $\Rightarrow$ bias (underfitting); too many $\Rightarrow$ variance (overfitting).
- ▶ Criteria differ only in the penalty weight c_T : $\hat{p}_{BIC} \leq \hat{p}_{HQC} \leq \hat{p}_{AIC}$.
- ▶ AIC is the **most lenient** and stays closest to the true lag order.
- ▶ Caveat: AIC is usually seen as prone to *overfitting*, but in a VAR each extra lag adds a full $(K \times K)$ block — so here even AIC can *underfit*.

Key insight (Ivanov & Kilian)

Recovering the *true* lag order is not the goal — minimizing MSE is. Sometimes an underfitting model wins.

Our first extension: BVAR shrinkage

The frequentist problem — curse of dimensionality:

- ▶ A 12-lag system needs $K^2p = 192$ coefficients against ~ 240 observations; each estimated coefficient costs one degree of freedom.

The shrinkage idea (Paper I): don't *discard* lags — *discipline* them by pulling coefficients toward a prior.

$$\text{posterior} \approx w (\text{data estimate}) + (1 - w) (\text{prior mean})$$

- ▶ The tightness λ_1 sets how hard we shrink.
- ▶ Keep all 12 lags, but rein in the higher-order coefficients.

What it solves

Shrinkage tames the **dimensionality** problem: many coefficients, few observations.

Our second extension: Bayesian model averaging

A different problem — model uncertainty:

- ▶ We do not know the true lag order. Committing to a single one ignores that uncertainty entirely.

The averaging idea (Paper II): keep *all* candidate lag orders and combine them.

$$p(\Delta \mid y) = \sum_k p(\Delta \mid M_k, y) p(M_k \mid y)$$

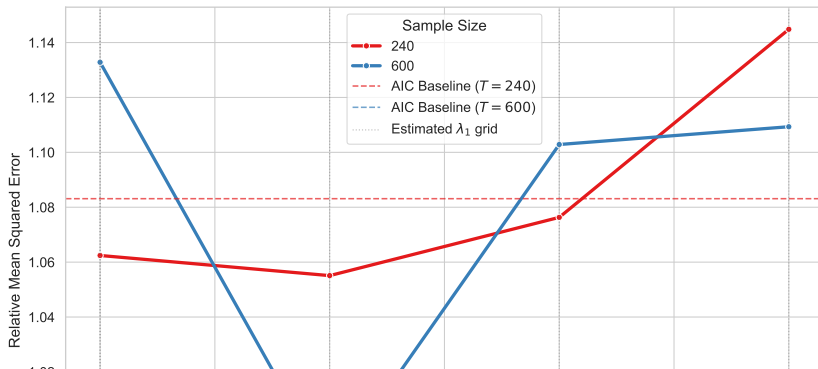
- ▶ Each model M_k (a lag order) is weighted by its posterior model probability.
- ▶ Instead of selecting one model, we **average** over many.

What it solves

Averaging addresses **model uncertainty**.

Simulation design in a nutshell

- ▶ A data-generating process is calibrated to the oil-market application of Kilian and Murphy (2014).
- ▶ VARs are estimated in repeated Monte Carlo samples.
- ▶ Structural IRFs are recovered using sign restrictions and the median target.
- ▶ Performance is evaluated by relative mean squared error.



Paper I: To Shrink or to Select?

- ▶ Focus: comparison of information criteria and Bayesian VAR shrinkage.
- ▶ Main methods: AIC, BIC, HQC, and BVAR with different shrinkage hyperparameters λ_1 .
- ▶ Main finding: AIC performs best among the classical criteria because it avoids severe underfitting.
- ▶ BVAR often outperforms AIC, especially when the true lag order is large.

Key results from Paper I

- ▶ BIC and HQC are too parsimonious in finite samples and introduce substantial bias.
- ▶ The BVAR approach reduces variance without relying on a hard lag truncation.
- ▶ A moderate prior shrinkage is typically more useful than very tight shrinkage.
- ▶ Forecast-oriented tuning, such as MDD-based selection, does not necessarily optimize structural IRF recovery.

Takeaway from Paper I

Conclusion

For structural inference, shrinkage can be more robust than selecting a single lag order when sample sizes are modest.

Paper II: Hedging Against Lag Order Uncertainty

- ▶ Focus: Bayesian model averaging as a strategy for dealing with model uncertainty.
- ▶ Main methods: averaging over lag orders and shrinkage specifications.
- ▶ Main finding: BMA reduces the sensitivity of IRF estimates to an uncertain lag order.
- ▶ The joint-grid BMA strategy often dominates the simple AIC benchmark.

Key results from Paper II

- ▶ Averaging across specifications diversifies uncertainty rather than forcing a single choice.
- ▶ BMA performs especially well when the true lag order is high and the data are noisy.
- ▶ The method protects against overfitting and underfitting at the same time.
- ▶ Its success rests on weighting a broad set of plausible models rather than selecting one.

Takeaway from Paper II

Conclusion

Model averaging is a natural response to lag-order uncertainty when the target is accurate structural inference.

How the two papers relate

Dimension	Paper I	Paper II
Main tool	BVAR shrinkage	Bayesian model averaging
Core idea	Regularize the VAR	Average over competing models
Advantage	Reduces variance	Hedging against model uncertainty
Main limitation	Prior choice matters	Computation and weighting matter

Shared conclusion

- ▶ In finite samples, hard lag selection can be too brittle.
- ▶ Shrinkage and averaging provide robust alternatives.
- ▶ The best strategy depends on the objective, but both papers support a more flexible treatment of lag uncertainty.

Concluding message

Bottom line

For set-identified SVARs, the most reliable inference often comes from methods that acknowledge uncertainty rather than forcing a single lag order.

- ▶ AIC remains a strong benchmark.
- ▶ BVAR and BMA are attractive alternatives when structural IRFs are the target.
- ▶ Moderate shrinkage and model averaging are especially promising in practice.

References

Ivanov and Kilian (2005), Kilian and Murphy (2014), Uhlig (2005), Fry and Pagan (2011), and the two papers written in this project.